

based on peripheral component interconnect (PCI) extensions for instrumentation (PXI) can benefit from widely available software and hardware components and easily connect with a PC to give high-performance results.

Industry's stringent requirements

If we categorise the applications further into aerospace and defence, these are more or less the same as in other domains like telecommunication, electrical safety testing, security and surveillance, and microwave and RF testing, but the sensitivity and tolerances required are very high.

The systems and components are designed and developed as per military standards and the test instruments are expected to measure as per the stringent tolerance levels and accuracies, and require calibration with standards.

Justifying this, S.R. Sabapathi, chief executive officer, QMAX Technologies, shares, "Typically, bonding testers used to test the aircraft body and instrument chassis resistance need to have accuracy of ± 0.001 ohm and calibration to the standards of NPL."

Semancik explains, "High accuracy and repeatability are key characteristics of these test equipment, making certain configuration elements essential, such as integrated signal conditioning, self-calibration, open transducer detection, noise immunity and stability. Some of the test environments can be quite demanding, exposing the instrumentation to high levels of acoustic noise, electromagnetic interferences, vibration and moisture. As a result, certain instruments must be sealed, conduction-cooled, splash-resistant and capable of withstanding extreme acoustic vibration."

"T&M instruments are used mostly in fully automated set-ups and not as standalone equipment. For example, testing of avionics equipment needs a test bench with all measuring instruments in the same bus to do full testing of voltage, frequency, power,



NI dual gigabit Ethernet interface

spectrum, etc," shares Sabapathi.

Avionics instruments used in aircrafts have to withstand extreme climatic conditions ranging from $+40^{\circ}\text{C}$ to -14°C (at 10,668 metres) and vibrations. So these need to be tested in simulated conditions at ground.

"For pneumatic and hydraulic systems used in aircraft controls, such as rudder, ailerons, wheels and brakes, we need special T&M instruments like data acquisition systems and other portable measuring instruments," says Sabapathi.

"It is necessary to have each test thoroughly documented and the data archived for future reference. This involves capturing instrumentation identification details and calibration records," adds Semancik.

General-purpose equipment for aerospace and defence

Most aerospace and defence systems and sub-systems are equivalent to commercial systems, so general-purpose T&M equipment satisfying the MIL standards are also used. Every change and advancement in technology is reflected in aerospace and defence missions.

"Remotely piloted vehicles—land, sea and air—scout ahead and stream back data and images. Secure wireless transmissions deliver voice, data and

images to war-fighters, who receive real-time intelligence about friends and foes. Advanced avionics and guidance systems ensure precise journeys for military and commercial aircrafts," expresses Awasthi.

The equipment, though general-purpose, need to be highly accurate to produce perfect real-time information. "The data acquired by the satellite for space station or radar for military intelligence—for analysing remote signals, detecting foreign signals, analysing and ensuring quality of content—should be lossless and real-time," says Narasimhan.

The high-technology aerospace industry is involved in the production of aircrafts, guided missiles, space vehicles, aircraft engines, propulsion units and related parts. "Signal generators, spectrum analysers, network analysers, EMI-EMC receivers and software, oscilloscopes, logic analysers, frequency counters, multimeters and power supplies are the general-purpose equipment required in design and development," shares Valsalan.

There are a variety of electronic and mechanical tests, such as structural test, functional test, EMI-EMC test, noise, vibration and harshness (NVH) test, and hardware in loop test, performed on the system. High-speed data acquisition apparatus is also an important component of T&M in aerospace and defence.

Cockpit instruments are tested on ground and air. Sabapathi adds, "Bonding tests are done to find the ground resistance between individual modules. It is required to be maintained to avoid static charge build-up on air while the aircraft is flying at a very high speed of around 833 km (450 nautical miles)."

Advanced automatic test equipment (ATE) with multiple test technologies are used to test new generation of multi-layered PCBs having high-density components like FPGAs and CPLDs in packages like BGA, micro BGA and SMD.